

DARANI KETTIMUTHU

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Work Authorization: H-4 EAD — Authorized to work in the USA through December 2028 · No sponsorship required

PROFESSIONAL SUMMARY

Results-driven IT professional and educator with 3+ years of experience training 200+ engineering students in Python, Java, SQL, HTML, Full-stack web development, RESTful API design, cloud-based AI systems, AWS Bedrock, RAG pipelines, and LLM integration. Holds an MS in Information Technology alongside a Google Cybersecurity Professional Certificate and AWS AI Practitioner credential. Proven track record of delivering deployed web applications, peer-reviewed publications, and AI-powered tools. Seeking to apply cybersecurity, cloud computing, Full Stack and AI skills in a dynamic technology environment.

CORE COMPETENCIES

Programming:	Python, Java, C, C++, HTML, CSS, JavaScript, SQL, Angular, React/Next.js, Database Design (SQL/NoSQL), MCP
Backend & APIs:	FastAPI (Python), RESTful APIs
DevOps:	Vercel Deployment, Docker, Kubernetes (k8s), AWS EKS, GHCR, GitHub Actions CI/CD
Cybersecurity:	Threat Detection, Phishing Analysis, Network Security, Google Cybersecurity Cert
Cloud & AI:	AWS Bedrock, LLM Integration, AWS EKS, RAG-style Retrieval, Agentic Workflows, AWS (AI Practitioner), IBM Cloud, AWS PartyRock, ChatGPT Prompt Engineering, TF-IDF Retrieval, Docker containerization, and Kubernetes orchestration.
Tools:	Android Studio, KEIL, Proteus, VS Code, Vercel
Soft Skills:	Technical Training, Curriculum Design, Code Review, Peer Review, Public Speaking

WORK EXPERIENCE

Software Engineer (Full Stack & QA) — *BadmintonScholar, Tamil Nadu, India* Feb 2021 – Sep 2022

- Built and enhanced full-stack web features using HTML, CSS, JavaScript, Python, improving performance and responsiveness across devices.
- Developed and integrated RESTful APIs with clear documentation; collaborated in Agile/Scrum sprints.
- Created and executed test cases, regression suites, and automation scripts, ensuring high quality releases.
- Gathered and translated business requirements into user stories and technical specs; supported UAT.
- Performed data modeling, database design, and SQL optimization for content and user data workflows.
- Maintained CI/CD pipelines using GitHub and improved deployment reliability through structured code reviews.
- Managed defects end-to-end using JIRA, reducing defect density and improving sprint delivery.
- Contributed to an analytics dashboard tracking KPIs for content performance and user engagement.

Placement Trainer — Technical & Soft Skills May 2018 – Feb 2021

Nandha Engineering College, Tamil Nadu, India

- Trained 200+ engineering students annually in Python, Java, C/C++, HTML, and SQL, contributing to improved campus placement rates by equipping students with industry-relevant skills.
- Designed and deployed 30+ coding exercises and assessments on GUVI digital platform, increasing student engagement and measurable skill completion rates.

- Conducted structured code reviews and unit-level testing sessions, reinforcing industry-standard debugging practices and reducing student assignment error rates.
- Prepared students for technical interviews through mock sessions, aptitude training, resume workshops, and hackathon guidance — resulting in multiple students securing placements at competitive companies.
- Mentored final-year project teams through the full project lifecycle — ideation, development, testing, and exhibition — at national-level college fests (VIGNANI 2K15 & 2K16).

Freelance Technical Reviewer & IEEE Volunteer

Jan 2024 – Present

[IEEE](#) | [IGI Global](#) | [NCWIT](#) | [EPICS in IEEE](#)

- Reviewed 10+ academic project proposals and conference papers for IEEE NMITCON, IEEE CASS, EPICS EAB, and IOP Publishing, ensuring high research quality and STEM education standards.
- Served as Judge at the 2025 NC Technology Student Association (NCTSA) Eastern Region Conference, evaluating student engineering projects against rigorous technical criteria.
- Proctored IEEEExtreme 18.0 (Oct 2024), supporting 24-hour global programming competition logistics for student participants across multiple teams.
- Recognized with Aspirations in Computing award by NCWIT (Feb 2025) for outstanding contributions to technology and community engagement.

PROJECTS

Smart Bookmark Manager — MCP Server | github.com/daranikettimuthu20/smart-bookmark-manager

- Built a local MCP server using Node.js and Anthropic MCP SDK exposing 4 custom tools to Claude Desktop for AI-powered bookmark management, tag-based organization, and live web page summarization via agentic tool chaining.
- Designed Zod-validated tool schemas, implemented SQLite CRUD operations using sql.js, and integrated cheerio and node-fetch for real-time HTML parsing and content extraction. **Tech: Node.js, MCP SDK, Zod, sql.js, cheerio, node-fetch, GitHub**

DocuChat AI — Multi-Format RAG Chatbot | docuchat-ai-w86t.vercel.app | github.com/daranikettimuthu20/docuchat-ai

- Built a production RAG chatbot supporting 7 file formats with automatic format detection, multi-turn memory, and a full custom RAG pipeline from scratch — no LangChain, no vector DB — including 500-word overlap chunking, TF-IDF relevance scoring, Top-K retrieval, and hallucination prevention via system prompt engineering. Powered by Claude Haiku API on Next.js 16.
- Containerized with a multi-stage Docker build (reducing image size by ~70%), orchestrated on Kubernetes with 2-replica deployment, rolling updates, and health probes — with a full GitHub Actions CI/CD pipeline auto-deploying to AWS EKS on every push to main via GHCR. Zero manual deployment steps. **Tech: Next.js 16, Claude Haiku API, TF-IDF Retrieval, unpdf, mammoth, xlsx, officeparser, Docker, Kubernetes, AWS EKS, GitHub Actions CI/CD, GHCR, Vercel**

H-4 EAD Career Navigator | h4ead-career-navigator.vercel.app

- Built an AI chatbot using Next.js, Claude API, TypeScript and Vercel helping 500K+ H-4 EAD holders navigate US job market and visa rules **Tech: Next.js, TypeScript, Tailwind CSS, Claude Haiku API, react-markdown, Vercel**

AI Resume Screener Web App | [AI Resume Screener Web App](#)

- Built and deployed an AI-powered Resume Screener using Next.js, Claude AI, and Vercel that performs ATS scoring, skill-gap detection, and cybersecurity analysis on uploaded resumes — enabling instant, data-driven job-readiness insights for users. **Tech: Next.js, Python/FastAPI, RESTful APIs, Claude API, RAG-style prompt routing, Vercel**

Personal Portfolio Website | [Darani Portfolio](#)

- Full-stack, AI-integrated professional portfolio built using Next.js, React, and modern UI/UX principles. Showcases deployed projects, publications, certifications, and AI tools with responsive design, optimized performance, and cloud-based hosting on Vercel. **Tech: Next.js, React, RESTful APIs, Python (FastAPI), database design, AWS Bedrock, LLM integration, RAG-style content retrieval, agentic workflows, GitHub CI/CD, Vercel deployment.**

Phishing Email Detection Tool | [Phishing-detector](#)

- Built and deployed a cybersecurity tool using HTML/CSS/JavaScript that classifies email content by risk level, detecting high-risk phishing patterns in real time — directly applying core threat-detection skills from Google Cybersecurity certification. **Tech: JavaScript, HTML/CSS, client-side ML patterns, threat-rule engine**

Blockchain-Based Certificate Verification System | [Capstone — University of the People, Dec 2025](#)

- Architected a decentralized credential verification system using blockchain technology to eliminate certificate forgery and enable tamper-proof, instant validation for academic institutions.

AI-Powered Smart Meal Planner | [partyrock.aws/u/Darani/6NEjohoOS](#)

- Developed a generative AI assistant using AWS PartyRock that produces personalized weekly meal plans based on dietary preferences and nutritional goals. **Tech: AWS Bedrock, generative AI workflows, LLM orchestration**

Smart To-Do List Web App | [to-do-list-psi-two-69.vercel.app](#)

- Engineered and deployed a full-stack task management app with real-time date, time, and geolocation features using JavaScript and Vercel continuous deployment. **Tech: JavaScript, Vercel, cloud functions, database design**

EDUCATION

MS — Information Technology (MSIT) | GPA: 3.83/4

Jan 2026

University of the People, Pasadena, CA, USA

M.E. — VLSI Designs | CGPA: 9.2 / 10

2020

Nandha Engineering College, Tamil Nadu, India

B.E. — Electronics & Communications Engineering | CGPA: 9.1 / 10

2018

Nandha Engineering College, Tamil Nadu, India

CERTIFICATIONS

- **Google Cybersecurity Professional Certificate** — Google / Coursera (Apr 2026)
- **Complete MCP Bootcamp – Build Next-Gen AI Agents with MCP** — Udemy (Apr 2026)
- **AWS AI Practitioner Challenge** — Udacity / AWS (Apr 2026)
- **Databricks Accredited Generative AI Fundamentals certification** — Databricks Academy (May 2026)
- **Claude 101, Claude Code 101, Claude Code in Action and Introduction to MCP** — Anthropic (Apr 2026)
- **AI+ Foundation** — AI CERTs® (Aug 2025)
- **PCEP — Certified Entry-Level Python Programmer** — OpenEDG Python Institute (Mar 2024)
- **Cloud Computing Fundamentals** — IBM SkillsBuild (Dec 2025)
- **ChatGPT Prompt Engineering for Developers** — DeepLearning.AI (Jan 2024)
- **Python Programming Language** — LinkedIn Learning (Feb 2026)
- **Foundations of Cybersecurity** — Google (Jan 2026)

PUBLICATIONS & RESEARCH

- Green Guard Intelligent Pesticide Spraying Solutions — OALib Journal, May 2025
- How AI Works: A Fun Guide for Kids — Amazon Publishing, Jun 2025
- The Time Travelling Lab Coats — Amazon Publishing, Oct 2025
- Feed Forward Pipelined Accumulate Unit for Machine Learning — IJSRED, Feb 2020
- College Bus Tracking Android Application Using GPS — IJIARE, Apr 2018

VOLUNTEERING & LEADERSHIP

- IEEE Professional & IEEE Young Professionals — active contributor to global engineering community and standards initiatives.

- EAB EPICS Grant Reviewer (IEEE) — evaluated engineering project proposals for real-world educational impact and funding merit.
- Reviewer — IGI Global, NCWIT IEEE Life Members Evolution Conference, IEEE NMITCON 2024, and IEEE Conference on Materials, Robotics, Automation, Computer & Control 2025.
- Effective Substitute Teacher — Wake County Public Schools, NC (Dec 2025), maintaining instructional continuity across technical and general education classrooms.